

Internal comprehensive audit of the H-LAYER -> T7 proof route (T-016..T-028):

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This note is a pre-enactment internal audit of the entire H-LAYER $\rightarrow$ T7 proof route. Its purpose is retraction-prevention: to re-verify, from start to finish, every step of the proof chain before the development is taken further, so that no later inconsistency forces a rollback. It introduces no new mathematical claim; it certifies the existing route. Five audit axes are applied:

1. **Numerical reproducibility** – re-run every verification script.
2. **Logical non-circularity** – each step’s hypotheses are met by priors; no cycle.
3. **Constant / convention consistency** – all load-bearing numbers from one source.
4. **Tier-claim honesty** – no over-claim; the canonical tier is respected.
5. **Superseded-version integrity** – pointers present; live content identified.

2. The proof route (logical DAG)

The route is a directed acyclic graph rooted at the canonical microscopic data (Math424_AddA: $u = -0.86$, $v = 3.24$, q_0 , the gap map $r_R(\mu^2)$, M_R):

$$\begin{array}{ccccccc}
 \underbrace{\text{T-016}}_{(D) \text{ iso}} & \rightarrow & \underbrace{\text{T-017}}_{(S) T' \leq 13} & \rightarrow & \underbrace{\text{T-018/9/20}}_{(O) R_{\text{lead}} < 1} & \rightarrow & \underbrace{\text{T-021}}_{(S) \text{ SC-SCOPE}} \rightarrow \\
 \underbrace{\text{T-022/3/4}}_{\text{competitor class}} & \rightarrow & \underbrace{\text{T-026}}_{\text{off-shell A,C}} & \rightarrow & \underbrace{\text{T-027}}_{\text{Blocker B}} & \rightarrow & \underbrace{\text{T-028}}_{\text{T7 assembly}} .
 \end{array}$$

Every edge points forward; no step depends on a later one. T-025 is a consolidation of T-016..T-024 and feeds nothing analytic.

3. Per-step verdict table

Step	Result (load-bearing)	Tier	Asserts
T-016	diagonal isotropy strict min; $\Phi''_{\text{diag}} = 4.03 > 0$, entropy $\geq \frac{1}{2}r_R^2$	T4	6/6
T-017	weighted Lemma A: $K_{\text{floor}} \leq T' \leq 13$ any amplitudes	STRONG EV.	3/3
T-018	DIAGONALITY LEMMA + $R_{\text{lead}}(13) = 0.650 < 1$ (live; b_{exch} superseded)	T3 red.	5/5
T-019	no A-independent Fock exchange (+ $\frac{3}{2}u_{\text{eff}}$ stabilising)	T4	5/5
T-020	2nd-cumulant off-diag stability class-wide ($\rho R_{\text{lead}} < 1$)	T4	5/5
T-021	SC-SCOPE joint $> 1 \iff T' < 60.4$; joint(13) = 1.097	T4	5/5
T-023	4 exclusion boundaries; $T' \leq N/2 \leq 20$ fallback	T4	5/5
T-024	operator two-class decision; kinetic penalty $\times 50$ MARGIN	DECISION	5/5
T-025	final consolidation (end-to-end)	CONSOL.	7/7
T-026	off-shell domination $\rho_{\text{off}}^{\text{ext}} = 0.572 < R_{\text{lead}} < 1$ (A,C)	T6-cand.	5/5
T-027	Blocker B: convention DERIVED, $R_{\text{max}} \leq 0.391$ enclosed, anchoring monotone	T6-cand.	5/5
T-028	T7 proposition assembled (non-circular)	T7-CAND.	5/5
Total			61/61

4. Axis 1 – numerical reproducibility

All twelve chain scripts (`res5_016..res5_028`) were re-executed from the canonical source; every assert passes (61/61). The audit kernel `res5_029_t7_route_audit.py` (5/5) re-derives the constants independently. No numerical claim in the route fails on re-run.

5. Axis 2 – logical non-circularity (the off-shell exclusion)

The single non-trivial circularity risk is the off-shell exclusion: it produces the registered $T' \leq 13$ class, so it must not assume that class. It does not. The NON-CIRCULAR form (T-028 v1.1) uses the adversarial $R_{\text{lead}}(20) = 0.974$, valid for ANY antipodal competitor before any reduction:

$$\rho_{\text{off}}^{\text{ext}} \leq R_{\text{lead}}(20) \frac{r_R}{r_R + c\delta^2} = 0.572 < 1.$$

Off-shell modes are thus strictly more stable for ANY competitor; only AFTER their exclusion do the survivors reduce to the on-shell registered class with $\rho_{\text{off}} = 0.381$, $R_{\text{lead}}(13) = 0.650$. The DAG (§2) confirms no other cycle.

6. Axis 3 – constant / convention consistency

Seventeen load-bearing constants were re-derived from `Math424_AddA` and matched the values quoted across the notes (worst $|\text{dev}|/\text{tol} = 0.49$):

$$\begin{array}{lll}
u = -0.86 & v = 3.24 & q_0 = 0.6802 \\
r_R = 0.3045 & u_{\text{eff}} = 2.685 & \Phi''_{\text{diag}} = 4.03 \\
\frac{1}{2}r_R^2 = 0.0464 & \text{const} = 23.2 & R_{\text{lead}}(13) = 0.650 \\
R_{\text{lead}}(20) = 0.974 & \text{joint}(13) = 1.097 & T'_{\text{crit}} = 60.4 \\
c\delta^2 = 0.214 & c\delta^2/\text{MARGIN} = 50 & \rho_{\text{off}} = 0.381 \\
\rho_{\text{off}}^{\text{ext}} = 0.572 & &
\end{array}$$

The load-bearing $\widehat{G^4}$ convention $1/(4\pi^2)$ is DERIVED (T-027 v1.1) from the $(2\pi)^3$ convolution measure (ϕ -integral = 2π); the Parseval ratio 1.0000 is a consequence; a $1/(2\pi^2)$ prefactor would double $\widehat{G^4}$ and give the spurious $R_{\text{max}} = 0.77$. The convention is thus not a free factor-2.

7. Axis 4 – tier-claim honesty

No current note claims an enacted T7. Every H-LAYER note carries one of "no tier flip", "T7-CANDIDATE / not enacted", or "B1/B2 T6 on {H-LAYER}". The strongest assembled status is T7-CANDIDATE (T-028). The legacy `res5-dr2-kappa-bound` "UNCONDITIONAL" wording was itself superseded (v1.2: "T7 modulo textbook NT"); it is not in the live route.

8. Axis 5 – superseded-version integrity

All eight notes re-issued $v1.0 \rightarrow v1.1$ (chi-link-bypass, gershgorin-threshold, isotropy-infimum-core, aadm-exclusion-boundaries, offshell-domination, blocker-b-hardening, hlayer-analytic-closure-consolidation, t7-proposition-assembly) carry the SUPERSEDED forward-pointer on $v1.0$. The live load-bearing content of T-018 is the DIAGONALITY LEMMA + $R_{\text{lead}} < 1$ (the gershgorin $v1.1$ carries a STATUS-UPDATE banner re-identifying the b_{exch} reduction as superseded by T-019); the T7 chain uses the live $R_{\text{lead}} < 1$, NOT the superseded b_{exch} phantom.

9. Residual register (what remains for T7)

Operator sign-off	required (no-auto-T7, GOVERNANCE.md)
CAS interval-arith. R_{max}	optional (enclosure already conservative)
selection $\text{joint}(13) = 1.097$	thin but > 1 , theorem-grade

No analytic gap remains. The two residual items are documentary/decisional, not mathematical blockers.

10. Honest scope (what the T7 decision does and does NOT close)

The T7 decision, if enacted, closes EXACTLY the H-LAYER / Reading-H / Proposition-A comparison theorem: G_* is the strict comparison infimum over $\mathcal{C}_{\text{phys}}$, discharging the H-LAYER hypothesis for B1-RH-ENUM and B2-PROPA-HLAYER. It does NOT prove TECT as a whole, nor complete the TOE: the Sector-C (Lorentz/GR), Sector-D (gauge/matter), and Sector-E/F (constants/cosmology) claims remain at their own tiers (and were tier-capped at T6 while GAP-1 was open; the cap lifts on enactment, but each still needs its own package).

11. Audit verdict

The H-LAYER $\rightarrow$ T7 proof route is internally consistent, reproducible, non-circular, and honestly tiered. All five axes pass. The route is ready for the operator T6 $\rightarrow$ T7 enactment decision; no rollback risk is identified in the audited chain. This audit enacts no tier change: B1/B2 T6 on {H-LAYER}, B5 T5.

12. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “An audit by the same author is not independent.” VALID, recorded: the audit re-derives constants from the canonical source independently of the note bodies and re-runs every script; but a fully independent referee pass (second author / external) remains advisable before publication. This is an INTERNAL audit, named as such.
- (β) “The selection margin 0.097 is thin – could a refined competitor breach it?” BOUNDED: the margin is on the registered on-shell $T' \leq 13$ class; off-shell and super- n_{pack} competitors are excluded by

theorem; within the class the joint is monotone in $\rho_{\text{lat}} = K_{\text{budget}}/(1 + T')$ and $T' \leq 13$ is pinned, so 0.097 is the worst case.

(γ) “Does the audit enact T7?” DISMISSED: no. It certifies readiness; the enactment is the operator’s. B1/B2 stay T6.

Quantitative sanity check (mandatory): *axis1* 61/61 asserts; *axis2* $\rho_{\text{off}}^{\text{ext}} = 0.572 < 1$ non-circular; *axis3* 17/17 constants (worst dev/tol 0.49); *axis4* 0 enacted-T7 claims; *axis5* 8/8 supersede pointers; *margins* 0.350, 0.097 > 0.

13. Result footer

Result ID:	B2-PROPA-HLAYER / H-LAYER->T7 route internal audit
Precise statement:	Pre-enactment 5-axis audit of T-016..T-028. (1) 61/61 script asserts re-run PASS; (2) off-shell exclusion non-circular ($\rho_{\text{off}}^{\text{ext}}=0.572<1$ via adversarial R_lead(20) before reduction); (3) 17/17 load-bearing constants re-derived from Math424 (worst dev/tol 0.49), the Ghat4 /(4 π^2) convention DERIVED from the (2 π) ³ measure; (4) no enacted-T7 over-claim; (5) 8/8 v1.0->v1.1 supersede pointers, live content identified (R_lead<1 live, b_exch superseded by T-019). Residual: operator sign-off + optional CAS R_max. Route internally consistent.
Scope:	The H-LAYER/Reading-H/Prop-A comparison theorem; operating endpoint I=2e-3; ROBUSTNESS-MU2 band. Does NOT cover Sectors C-F or the TOE.
Dependencies:	the full chain T-016..T-028 (each note + script); res5_029_t7_route_ audit.py (constant kernel); Math424_AddA (canonical source).
Evidence grade:	AUDIT (re-derivation + re-run) + EXECUTED (res5_029 5/5 + 12 chain scripts 61 asserts).
Reproduction command:	python codes/vacuum/res5_029_t7_route_audit.py (+ the 12 chain scripts)
Expected output:	17/17 constants consistent; margins 0.350/0.097>0; 61/61 chain; 5/5.
Falsification gate:	Any chain script failing on re-run; a constant inconsistent with Math424; a logical cycle; an enacted-T7 over-claim; a missing supersede pointer; or a competitor in C_phys with $F[Q] \leq F[G_*]$.
Tier before / after:	No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. The route is certified ready for the operator T6->T7 enactment decision; no rollback risk found.
No-overclaim statement:	This is an INTERNAL audit, not an independent referee pass, and it does NOT enact T7. It certifies the route’s internal consistency and

reproducibility. The selection margin (0.097)
is thin but theorem-grade.
A second/external review is advisable before
publication. B1/B2 stay T6.

Next required action: OPERATOR: (i) optional external/second-author
referee pass; (ii) the
T6->T7 enactment sign-off; (iii) optional CAS
interval $R_{\max}$ tightening.